WORKSHEET 8.9

Solids of Revolution: Washers

Unit 8: Applications of Integration

Key ideas

- If the region does not touch the axis, each slice is a washer: $V = \pi \int_a^b (R^2 - r^2)\,dx$.
- R is the distance from the axis to the **far** boundary; r is the distance to the **near** boundary.
- Square each radius separately. $(R - r)^2$ is wrong.
- About $y = k$, both radii are distances from k: for a line below the region, $R = \text{top} - k$ and $r = \text{bottom} - k$.

Worked examples

Example 1 · About the x-axis

The region enclosed by $y = 2x$ and $y = x^2$ is revolved about the x-axis. Find the volume.

Solution. They meet at $x = 0, 2$; $R = 2x$, $r = x^2$: $\pi \int_0^2 (4x^2 - x^4)\,dx$.

$\frac{64\pi}{15}$.

Example 2 · About the y-axis

The same region is revolved about the y-axis. Find the volume.

Solution. $R = \sqrt{y}$, $r = \frac{y}{2}$, $0 \le y \le 4$: $\pi \int_0^4 \left(y - \frac{y^2}{4}\right)\,dy$.

$\frac{8\pi}{3}$.

Example 3 · About a line below

The region bounded by $y = \sqrt{x}$, the x-axis and $x = 4$ is revolved about the line $y = -1$. Find the volume.

Solution. $R = \sqrt{x} + 1$, $r = 1$: $\pi \int_0^4 (x + 2\sqrt{x})\,dx$.

$\frac{56\pi}{3}$.

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Practice

CORE

1. Find the area of a washer with outer radius 5 and inner radius 3.

2. The region between $y = 3$ and $y = 1$ for $0 \leq x \leq 2$ is revolved about the x-axis. Find the volume.

3. Write, but do not evaluate, an integral for the volume when the region enclosed by $y = \sqrt{x}$ and $y = x$ is revolved about the x-axis.

4. Evaluate the integral in Question 3.

EXAM LEVEL

5. The region enclosed by $y = x^2$ and $y = 4$ is revolved about the x-axis. Find the volume.

6. The region in the first quadrant enclosed by $y = x^3$ and $y = x$ is revolved about the x-axis. Find the volume.

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7. The region enclosed by $y = x^2$ and $y = x$ is revolved about the line $y = 3$. Find the volume.

8. *Calculator.* The region bounded by $y = e^x$, the y-axis and $y = 3$ is revolved about the x-axis. Find the volume to three decimal places.

Question 9 · Free-response practice

Let R be the region enclosed by $y = x^2 + 1$ and $y = 2x + 1$.

(a) Find the area of R.

(b) Find the volume of the solid generated when R is revolved about the x-axis.

(c) Write, but do not evaluate, an integral for the volume of the solid generated when R is revolved about the line $y = -2$.

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Worked solutions

1. $\pi(25-9)=16\pi$.

2. $\pi(9-1)(2)=16\pi$.

3. $\pi\displaystyle\int_0^1 (x-x^2)\,dx$.

4. $\pi\left(\dfrac{1}{2}-\dfrac{1}{3}\right)=\dfrac{\pi}{6}$.

5. $\pi\displaystyle\int_{-2}^{2}(16-x^4)\,dx=2\pi\left(32-\frac{32}{5}\right)=\frac{256\pi}{5}$.

6. $\pi\displaystyle\int_0^1 (x^2-x^6)\,dx=\pi\left(\frac{1}{3}-\frac{1}{7}\right)=\frac{4\pi}{21}$.

7. $R=3-x^2$, $r=3-x$: $\pi\displaystyle\int_0^1 (x^4-7x^2+6x)\,dx=\pi\left(\frac{1}{5}-\frac{7}{3}+3\right)=\frac{13\pi}{15}$.

8. $\pi\displaystyle\int_0^{\ln 3}(9-e^{2x})\,dx=\pi(9\ln 3-4)\approx 18.496$.

9. (a) The graphs meet at $x=0$ and $x=2$: $\displaystyle\int_0^2 (2x-x^2)\,dx=\frac{4}{3}$.

   (b) $\pi\displaystyle\int_0^2\left[(2x+1)^2-(x^2+1)^2\right]dx=\pi\int_0^2(-x^4+2x^2+4x)\,dx=\frac{104\pi}{15}$.

   (c) $\pi\displaystyle\int_0^2\left[(2x+3)^2-(x^2+3)^2\right]dx$.

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